

Avalux: An Open Integrated Platform for Moderated Usability Testing with Real-Time Evaluator–Participant Synchronization

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Abstract—Moderated usability testing remains the reference method for discovering interaction problems, yet its instrumentation is fragmented: evaluators juggle stopwatches, spreadsheets, questionnaire tools, and manual report assembly, while commercial platforms focus on unmoderated remote studies and treat moderated sessions as video to be coded afterwards. We present *Avalux*, an open, self-hostable web platform that integrates the complete moderated-testing pipeline: reusable test protocols (tasks with optimal-path baselines, typed error taxonomies, per-task questions including in-browser audio/video/photo capture, interview scripts, and study-specific participant attributes); a live evaluator cockpit that logs per-task completion time, action counts, categorized errors, hesitation events, and Single Ease Question ratings; and a participant client that runs on the participant’s own device via a join link, synchronized with the evaluator in real time over WebSocket channels, including a gating protocol that prevents the session from advancing while the participant is still answering. Sessions conclude with an automatically scored System Usability Scale questionnaire and a semi-structured interview, and export to a self-contained PDF report. We describe the architecture—a React single-page application over a Postgres backend with row-level security—and report two case studies: a four-participant evaluation of a household touch-panel controller (mean SUS 28.1) and a four-participant evaluation of an eye-tracking analysis dashboard (mean SUS 80), demonstrating that the platform supports end-to-end studies and discriminates sharply between designs of different quality.

Index Terms—usability testing, moderated evaluation, SUS, SEQ, research tools, human–computer interaction

I. INTRODUCTION

Task-based moderated testing with a small number of participants remains the workhorse of usability practice: a facilitator observes a participant attempting representative tasks, records performance and difficulties, and complements observation with standardized instruments such as the System Usability Scale (SUS) [1] and the Single Ease Question (SEQ) [2]. Classic problem-discovery models show that even four to five participants uncover the majority of severe problems [3], [4], which keeps the method attractive for small teams and academic projects.

The tooling situation is less attractive. Commercial platforms concentrate on unmoderated remote studies, where metrics are captured by automatic instrumentation of the participant’s browser; their moderated modes produce session

videos and free-form notes that must be coded after the fact. In practice, a moderated session is therefore run with an ad-hoc toolchain—a stopwatch, a paper or spreadsheet observation grid, a separate questionnaire form, and manual report writing—which is slow, error-prone, and hard to standardize across studies.

This paper presents *Avalux*, an open web platform that integrates the moderated-testing pipeline end to end. Its contributions are:

- an **integrated protocol model**: reusable templates couple tasks (with optimal time and action-count baselines) to typed error taxonomies, per-task questions (open, choice, rating, and in-browser audio/video/photo capture), interview scripts, and study-specific participant attributes;
- a **live moderated session** design in which the evaluator’s cockpit (timer, action counter, one-tap typed error and hesitation logging, per-task SEQ) and the participant’s own device are synchronized in real time, including a gating protocol that blocks task advancement while the participant is still answering the previous task’s questions;
- **two case studies**—a household touch-panel controller and an eye-tracking analysis dashboard—run entirely on the platform, showing that it supports realistic studies and yields sharply discriminating results (mean SUS 28.1 vs. 80).

Avalux is deployed at <https://avalux.pt>; its schema, protocol seeds, and report generator are versioned with the application source.

II. RELATED WORK

A mature market of commercial software-as-a-service platforms now supports remote usability evaluation. UserTesting combines a proprietary participant panel with video-first moderated (“Live Conversation”) and unmoderated studies overlaid with AI-assisted analysis [5], and has absorbed the quantitative benchmarking platform UserZoom following their 2023 merger [6]. Maze is unmoderated by default, converting prototypes and live websites into task-based studies scored through automatically instrumented metrics such as completion rate, misclick rate, and task duration [7]. Lookback specializes in live moderated sessions with screen, camera, and audio

capture, stakeholder observation rooms, and timestamped free-form notes [8]; Loop11 supports both modes and, unusually, auto-computes SUS and NPS for unmoderated tests [9]; Useberry targets prototype-centric unmoderated testing with click and flow analytics [10]. Across these tools, quantitative evidence derives almost entirely from automated client-side instrumentation in unmoderated studies, while moderated modes yield video recordings and free-form notes that must be coded after the fact. To the best of our knowledge, none offers a structured real-time evaluator-logging instrument for moderated sessions, per-task ease ratings integrated into a moderated pipeline, or an open, self-hostable architecture.

Academic work on remote evaluation dates to Hartson et al., who framed the network as an extension of the usability laboratory [11]. Andreasen et al. subsequently found remote synchronous testing “virtually equivalent” to conventional lab-based think-aloud, whereas asynchronous methods identify fewer problems at lower moderator cost [12], an asymmetry confirmed by Bruun et al. [13] and persisting in recent comparisons [14]. On the tooling side, RemUSINE and WebRemUSINE pioneered task-level instrumentation by comparing interaction logs against task models, but operate post hoc rather than steering live sessions [15], [16]. TechSmith’s Morae became the de facto observation suite in usability practice [17], [18], yet it is a discontinued desktop product requiring installed software on the participant’s machine. More recent browser-based platforms such as UTAssistant [19], [20] and the analytics-augmented test-management system of Generosi et al. [21] target largely asynchronous workflows, and an action-research deployment of remote testing found the evaluation tool itself to be an obstacle to high-quality results [22].

Avalux’s measurement model follows the ISO 9241-11 decomposition of usability into effectiveness, efficiency, and satisfaction [23]. Perceived usability is captured with the ten-item SUS [1], whose reliability [24], [25], benchmark mean of 68 [26], and adjective-based interpretation [27] are well established; per-task difficulty uses the Single Ease Question, shown to perform comparably to more elaborate post-task instruments [2]. Completion rate, time-on-task, and action counts relative to an optimal path are the standard objective performance metrics [28], [29], and moderated testing with small samples is supported by classic problem-discovery models [3], [4].

A final thread concerns where sessions take place and on whose device. Early comparisons found no significant remote-versus-local differences in critical errors or time-on-task [30], [31], while lab-versus-field studies of mobile applications show that removing participants from their natural device context can mask context-dependent problems [32], [33]. The COVID-19 pandemic pushed moderated protocols onto general-purpose videoconferencing [34], [35], with studies of remote moderation identifying the moderator’s reduced visibility into the participant’s device and context as its principal cost [36].

Avalux addresses the gap these threads leave open. It is an

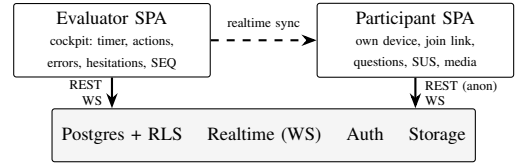


Fig. 1. Architecture: two browser roles over a shared Postgres backend; row-level security scopes evaluator data to accounts and participant access to per-session join codes; realtime channels synchronize both views.

open, integrated platform for moderated testing in which the evaluator live-logs, per task, completion time, action counts, typed errors, hesitation events, and SEQ ratings, while the participant completes tasks on their own device via a browser join link, with task state synchronized between the two roles in real time over WebSockets. A single pipeline carries a session from protocol definition through per-task SEQ, post-session SUS, and a semi-structured interview to an automatically generated PDF report. Unlike the commercial services surveyed above, Avalux is open and self-hostable—a React single-page application over Supabase Postgres with row-level security and realtime channels—and unlike prior academic tooling it makes fine-grained evaluator-side logging and evaluator-participant realtime synchronization first-class features of the moderated session rather than post hoc analyses.

III. SYSTEM DESIGN

A. Architecture

Avalux is a React single-page application (TanStack Router, Tailwind, Recharts) served statically from a CDN, with all persistent state in a hosted Postgres instance accessed through PostgREST and secured by row-level security (RLS) policies (Fig. 1). Three backend primitives carry the design: (i) *RLS dual-role security*: evaluator rows are scoped to their authenticated account, while participants operate anonymously, authorized per session through unguessable join codes checked inside the policies themselves; (ii) *realtime channels*: both clients subscribe to Postgres change streams over WebSockets, so every mutation (task advanced, answer submitted) propagates to the other role without polling; (iii) *storage buckets* hold media answers captured in the participant’s browser via `getUserMedia`. There is no custom application server, which keeps the platform self-hostable on free tiers of commodity services.

B. Protocol Model

A *template* is a reusable test protocol: an ordered set of *tasks*, optionally grouped, each carrying instructions and optimal time/action baselines; a study-specific *error taxonomy* (code plus label, e.g. *WB—wrong button*); *per-task questions* of seven types (open text, single and multiple choice, rating, audio, video, photo); a semi-structured *interview script*; and *participant fields*—extra attributes such as handedness or vision correction that the study collects per participant. A *session* instantiates a template for one participant and accumulates per-task results (status, time, action count, error and hesitation logs

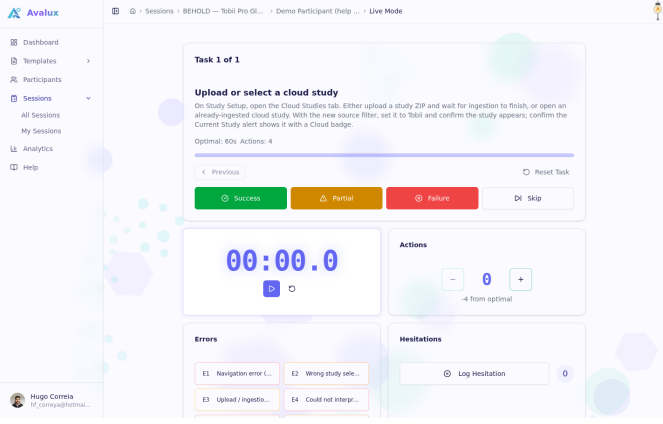


Fig. 2. Evaluator cockpit during a live session: task navigator with completion controls, timer, action counter, one-tap typed error logging, and hesitation logging.

with timestamps, SEQ), question answers, interview answers, and the ten SUS item scores. Participants can be enrolled directly, through single-use personal links, or through shared links that spawn one session per respondent, with configurable collection (or omission, for anonymous participation) of identity fields.

C. Live Session and Synchronization

During a session the evaluator’s cockpit drives the protocol (Fig. 2). Completing a task opens the SEQ prompt and advances a persisted task pointer; the participant’s device follows that pointer and shows the active task’s instructions, so the facilitator never has to read them aloud. When the evaluator marks a task complete, its questions unlock on the participant’s device; conversely, the evaluator’s success/partial/failure controls are *disabled* while the participant is still answering the previous task’s questions, with a notification when they finish—a two-way gate that keeps both roles aligned without verbal coordination. The evaluator can step back one task without destroying recorded data, or explicitly reset a task (clearing its metrics, logs, and answers) after confirmation. After the last task the participant completes the interview and the SUS questionnaire on their own device; scores are computed and banded automatically [27], [26].

D. Security Model

The platform’s least conventional engineering choice is placing the whole authorization model inside the database. Evaluator tables carry policies of the form `user_id = auth.uid()`; participant access is granted to the anonymous role through policies that join the target row to a session whose unguessable `join_code` is non-null, so possession of a link is the capability to act in exactly one session. Because participants must also *write* (answers, SUS items, media references), each such table pairs a scoped `INSERT/UPDATE` policy with the read policy, and mutable invitation counters are further restricted with column-level grants so an anonymous client can increment a response counter but not rewrite an



Fig. 3. Session results dashboard: per-task outcomes with time, action, error, hesitation, and SEQ columns; charts below compare performance against the template’s optimal baselines.

invitation’s task selection. This model survived two studies without an application server, though it requires discipline: one gap—a missing anonymous update policy on invitation counters, which silently aborted link joins after the participant row was already created—was found during operation and is now covered by a regression migration. Studies that must avoid personal data can disable identity collection entirely, producing anonymous participant records that never join the evaluator’s roster.

E. Analysis and Reporting

Session dashboards chart completion status, time and action counts against the template’s optimal baselines, error and hesitation distributions, SEQ per task, and a normalized per-task performance radar (Fig. 3); template-level analytics aggregate all completed sessions. A one-click export renders a self-contained PDF report (participant demographics, per-task tables, charts, logs, interview answers, and SUS breakdown with cross-session comparison), replacing the manual assembly step that typically follows moderated studies.

IV. CASE STUDIES

Two studies were run end to end on the platform. Both used in-person moderated sessions in which the participant handled the system under test while the evaluator logged performance in Avalux; study 2 additionally used the participant’s own device for questions, interview, and SUS via join links.

A. Study 1: Household Touch-Panel Controller

Four participants (ages 20–23, two female, three of low and one of high self-rated technical proficiency) attempted 14 tasks—seven simple (e.g. wake the panel, read the time, adjust temperature) and seven complex (e.g. set the clock, change language, recover from an error state)—on a commercial water-heater touch panel. Sessions averaged 15 minutes. Aggregate completion was 70% success, 20% partial, and 11% failure. SUS scores were 25.0, 40.0, 17.5, and 30.0 (mean 28.1, “Poor” [27], far below the benchmark mean of 68 [26]);

TABLE I
STUDY 1: SELECTED TASK-LEVEL RESULTS ($N=4$).

Task	Mean SEQ	Mean time (s)	Note
Wake up panel	1.8	—	all participants struggled
Recover from error	2.5	—	no error feedback shown
Set clock to 08:30	—	43.6	10 actions vs. 8 optimal
Read current time	7.0	2.7	100% success

notably the high-proficiency participant produced the *lowest* score, indicating problems not attributable to inexperience. Task-level metrics located the failures precisely (Table I): the undiscoverable wake-up gesture (SEQ 1.8, all participants struggled), error recovery without any error feedback (SEQ 2.5, participants resorted to power cycling), and clock adjustment (43.6 s and 10 actions against an optimal 8). The typed error taxonomy attributed most errors to inconsistent button semantics across screens, which became the study’s principal design recommendation.

B. Study 2: Eye-Tracking Analysis Dashboard

The second study evaluated BEHOLD, a post-hoc analysis dashboard for Tobii Pro Glasses 3 recordings, with four participants (ages 21–23, students, low technical proficiency). The protocol exercised the platform’s richer features: 20 tasks in eight groups following the analyst’s journey (study ingestion, data browsing, areas of interest, times of interest, visualizations, participant comparison, metrics exploration, export), a 19-item error taxonomy, media question types (screenshots, screen recordings, audio narration), and custom participant fields (handedness, vision correction, device, location). Task success was near-universal (96% of task attempts) with one consistent exception: configuring a fixation metric in the metrics explorer failed for participants across the cohort, isolating the dashboard’s least intuitive workflow. SUS scores were 80, 80, 82.5, and 77.5 (mean 80, “Good”, approaching “Excellent” [27]). Open answers converged on three concrete defects—an auto-track feature hidden in an overflow menu, participant IDs shown instead of names in comparisons, and truncated axis labels—each traceable to a specific task’s logs and answers in the report.

C. Cross-Study Reading

Fig. 4 contrasts the per-participant SUS scores of both studies against the industry benchmark. The two cohorts are comparable (students in their early twenties, predominantly low technical proficiency), yet the platform separates the two designs by more than fifty SUS points with non-overlapping ranges, and its task-level instruments explain *where* each design earns its score. We take this as evidence that an integrated moderated pipeline can deliver both the discriminative summary metrics and the diagnostic detail that typically require separate tools.

Operationally, a single evaluator ran every session in both studies. The protocol-as-template design meant the second study’s considerably richer instrumentation (19 error codes,

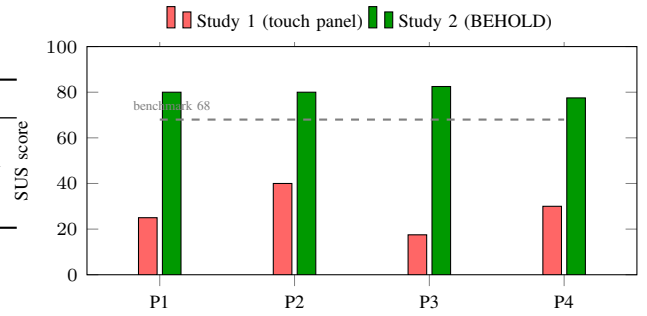


Fig. 4. Per-participant SUS scores in both case studies against the industry benchmark mean of 68 [26].

media questions, custom participant attributes) cost setup effort once rather than per session, and both study reports—including all charts and per-task appendices—were generated without manual assembly. The same integration surfaced platform defects the studies would otherwise have hidden: participants’ own difficulty reports (truncated chart labels, identifier-only comparisons) doubled as usability feedback *on the testing platform itself* and were fixed between studies.

V. DISCUSSION AND LIMITATIONS

Both studies used four participants—adequate for problem discovery [3], [4] but not for statistical generalization of the metric values. Action and error counts are logged by the evaluator rather than instrumented automatically; this keeps the platform applicable to physical devices (study 1 evaluated hardware no browser can instrument) at the cost of observer load and potential undercounting during fast interaction bursts. SEQ and SUS are self-report instruments with known ceiling effects. We have not yet run a controlled comparison of evaluator effort against an ad-hoc toolchain or a commercial moderated platform; the case studies demonstrate feasibility and output quality, not superiority. Finally, the anonymous-participant security model trades open link joins for RLS policies keyed on unguessable codes; a formal audit of that policy surface (one gap in invitation-counter updates was found and fixed during study operation) is ongoing work.

A further practical consideration is where a hybrid design like this sits between the lab and the fully remote setting. Because the participant client is an ordinary web page, the same protocol runs in a lab (evaluator and participant co-located, as in both case studies), over a video call with the join link shared in chat, or asynchronously through shared links when moderation is not required—though the platform’s distinctive instruments (evaluator-logged actions, errors, and hesitations) only apply in the moderated modes. Prior comparisons suggest synchronous remote moderation approaches lab quality [12], [31]; verifying that Avalux’s gating protocol preserves that equivalence when the moderator is not physically present is a natural next experiment. Finally, the participant-own-device model inherits the device diversity of field settings [33]: the mobile-first participant client was a direct consequence of pilot participants joining on their phones.

VI. CONCLUSION AND FUTURE WORK

Avalux integrates protocol definition, live moderated logging, participant-side questioning on the participant's own device, standardized instruments, analytics, and reporting into one open platform, and carried two complete usability studies whose results discriminated sharply between a poorly and a well-received design. Planned work includes a more flexible task model (removing the fixed simple/complex dichotomy), per-session data anonymization for privacy-sensitive studies, optional automatic event capture when the system under test is browser-based, and multi-evaluator observation of a single live session.

AVAILABILITY

The platform is deployed at <https://avalux.pt>. Its database schema (31 SQL migrations), the seeded protocol of study 2, the in-application tutorial, and the report generator are versioned together with the application source.

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